

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 10

Moment Inequality

Theorem: Let X be a RV and $g : [0, \infty) \rightarrow [0, \infty)$ be a non-decreasing function such that $E(g(|X|))$ is finite. Then for any $c > 0$ with $g(c) > 0$, then

$$P(|X| \geq c) \leq \frac{E(g(|X|))}{g(c)}.$$

Corollary: (Markov Inequality) Let X be a RV with $E(|X|^r) < \infty$ for some $r > 0$. Then for any $c > 0$,

$$P(|X| \geq c) \leq \frac{E(|X|^r)}{c^r}.$$

Corollary: (Chebyshev Inequality) Let X be a RV with $E(X^2) < \infty$. Let us denote $\mu = E(X)$ and $\sigma^2 = \text{Var}(X)$. Then for any $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Example 1: (Chebyshev inequality is tight) Let X be a DRV with PMF

$$f_X(x) = \begin{cases} \frac{1}{8} & \text{if } x = -1, 1 \\ \frac{3}{4} & \text{if } x = 0 \\ 0 & \text{otherwise.} \end{cases}$$

Then $E(X) = 0$ and $E(X^2) = \frac{1}{4}$.

Using Chebyshev inequality, $P(|X| \geq 1) \leq \frac{1}{4}$.

Using PMF, $P(|X| \geq 1) = \frac{1}{4}$.

Gamma Function

Def: For $\alpha > 0$, define

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt.$$

Theorem:

- ① The functional equation $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$ holds for $\alpha > 0$.
- ② $\Gamma(n + 1) = n!$ for $n = 1, 2, \dots$

Beta Function

Def: For $\alpha > 0$ and $\beta > 0$, define

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

Theorem: For $\alpha > 0$ and $\beta > 0$,

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Corollary: $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$

Gamma Distribution

A RV X is said to have a gamma distribution if the PDF of the RV is given by

$$f_X(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} & \text{if } x > 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$. We will use the notation $X \sim \text{Gamma}(\alpha, \beta)$.

Remark: $E(X) = \frac{\alpha}{\beta}$, $\text{Var}(X) = \frac{\alpha}{\beta^2}$.

Remark: $M_X(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha}$ for $t < \beta$.

Beta Distribution

A RV X is said to have a beta distribution if the PDF of the RV is given by

$$f_X(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$. We will use the notation $X \sim \text{Beta}(\alpha, \beta)$.

Remark: $E(X) = \frac{\alpha}{\alpha+\beta}$, $\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.